PHYSICAL CHEMISTRY CUMULATIVE EXAM

April 4, 1998

Sketch the following molecules and determine their symmetry point group. Indicate if the molecule may have a dipole moment or optical activity.

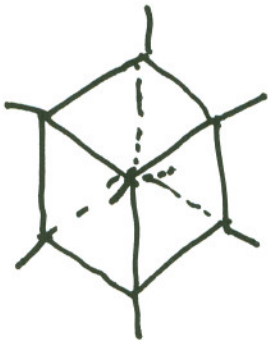
(a) hydrazine H_2N -NH_2 (gauche configuration)

(b) ethylene

(c) methyl amine

(d) Buckminster fullerene, C_{60} (no sketch needed)

(e) cubane C_8H_8



(f) $Ni(CN)_4^{2-}$ (planar)

(g) $Fe(CO)_5$ (trigonal bipyramid)

(h) BF_3

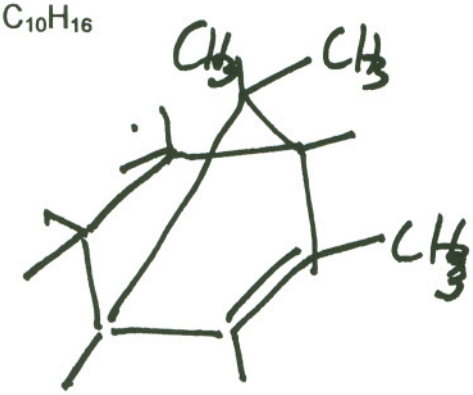
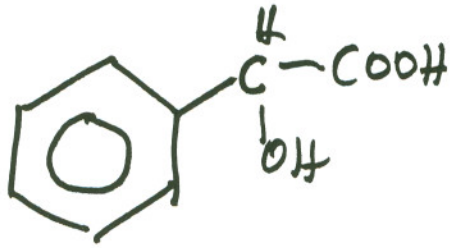
(i) BF_4^-

(j) p-benzoquinone, $C_6H_4O_2$

(k) norbornadiene, C_7H_8

(l) NF_3 (pyramidal)

(m) α-pinene, $C_{10}H_{16}$



(n) mandelic acid